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**European Patent Application No. 22755628.9**
**P2 Holding Oy**

We refer to the official communication pursuant to Rules 70(2) and 70a(2) EPC dated 21 January 2025 and confirm that the applicant wishes to proceed further with the above European Patent Application.

Further, we refer to the extended European search report dated 3 January 2025 and wish to submit the following in support of the above European Patent Application.

**Amendments**

The enclosed amended set of claims 1 – 13 replaces the claims 1 – 13 presently on file.

Thus, the amended independent claim 1 discloses (amended features underlined):
A computer implemented method, comprising:

obtaining a previously captured image data of a virtual tour of a location,

receiving real-time image data of a target object (101, 102) located within a field of view area (106, 107) of a camera (103, 104), which target object (101, 102) is located in the same location wherein the previously captured image data was captured,

analysing the real-time image data for detecting a predetermined deviation, and

overlaying the real-time image data of the target object (101, 102) on a corresponding area of the target object (101, 102) in a virtual tour view for forming a mixed image virtual tour data, only if the predetermined deviation is detected, otherwise the virtual tour view of the target object (101, 102) is shown,

receiving real-time acoustic data captured by an audio sensor (201d) in the location,

analysing the received acoustic data for detecting a malfunction, and

**characterized** in that the method further comprises:

playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor (201d) or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data.

"predetermined deviation" (the amended feature) has been disclosed in support e.g. on page 17 lines 28-33 in the description filed in PCT phase, the rest of the supporting parts reference also to the same PCT document as filed).

Claim 3 is amended as follows: 3. A computer implemented method according to claim 1 or 2, wherein the previously captured image data comprises images captured by a ~~person walking~~ moving platform moving through the location, and which previously captured image data is obtained from a camera, a memory or a cloud storage, and wherein the method further comprises: constructing the previously captured image data as a virtual tour by a data processing unit. Support for this amendment can be found on page 7 lines 17-20.

Claim 4 is amended as follows: 4. A computer implemented method according to claim 1 or 2, wherein the previously captured image data is a ready-constructed virtual tour made from video image data or still images. Support for this amendment can be found on page 7 lines 20-30. Static images or video of areas along a real-world virtual tour environment must be constructed to a virtual tour

Claim 8 is amended as follows: 8. A computer implemented method according to any of the claims 1 to 7, wherein a time point of ~~a~~ views of the mixed image virtual tour is changeable to a previous time point. Support for this amendment can be found on page 15 line 36- page 16 line 11. Claim 12 is amended correspondingly.

Independent claim 9 is amended correspondingly as claim 1.

Thus, all the features of the new claims are disclosed in the application as filed. Hence, the amended claims comply with the Art. 123(2) EPC.

The description is be brought into conformity with new claims, where applicable.

D1 – D4 are mentioned and briefly discussed in the description.

Reference numbers were mainly already added to the claims, but a few more have now been added.

Clarity
The previously presented amendments clarify the claims 1 and 4. Thus, the claims are clear according to Art. 84 EPC.

**Novelty (Art. 54 EPC)**

The amended independent claim 1 discloses: A computer implemented method, comprising:

obtaining a previously captured image data of a virtual tour of a location,

receiving real-time image data of a target object (101, 102) located within a field of view area (106, 107) of a camera (103, 104), which target object (101, 102) is located in the same location wherein the previously captured image data was captured,

analysing the real-time image data for detecting a predetermined deviation, and

overlaying the real-time image data of the target object (101, 102) on a corresponding area of the target object (101, 102) in a virtual tour view for forming a mixed image virtual tour data, only if the predetermined deviation is detected, otherwise the virtual tour view of the target object (101, 102) is shown,

receiving real-time acoustic data captured by an audio sensor (201d) in the location,

analysing the received acoustic data for detecting a malfunction, and

**characterized** in that the method further comprises:

playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor (201d) or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data.

D1 fails at least to disclose: analysing the real-time image data for detecting a deviation, and the overlay is performed only if the deviation is detected, and
receiving real-time acoustic data captured by an audio sensor in the location,
analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is detected for further augmenting the mixed image virtual tour by real-time acoustic data.

Thus the amended claim 1 is novel in view of the D1 in accordance to Art. 54(1) EPC.

D2-D4 also fails to disclose all the features of claim 1, as stated in the OA.

Thus the amended claim 1 is novel in view of the D2-D4 in accordance to Art. 54(1) EPC.

The same analogy applies to the independent claims 9 and 13 as well. Thus, the independent claims 1, 9 and 13 are novel in sense of Art. 54(1) EPC.

**Inventive step (Art. 56 EPC)**

D1 is regarded as the closest prior art in the official communication.

The following was stated in the OA:
"*The subject-matter of claim 1 differs in that it is further claimed*

*1) receiving real-time acoustic data captured by an audio sensor in the location, analysing the received acoustic data for detecting a malfunction, playing the real-time acoustic data in an area of the mixed image virtual tour view corresponding to the location of the audio sensor or its surroundings in the location if the malfunction is*

*detected for further augmenting the mixed image virtual tour by real-time acoustic data.*

*2) analysing the real-time image data for detecting a deviation, and the overlay is performed only if the deviation is detected.*

*It is noted that the above differences solve two different technical problems*

*The first difference solves the problem of improving the virtual tour in order to render it more realistic*

*The second solves the problem of how reducing the computational load (and the transmission band) of the system due to the overlay of the real time images.*

*Due to this, the two features do not have a synergetic effect and render the claim a mere juxtaposition of features that do not concur to solve the same technical problem (Guidelines G-VII.7).".*

The applicant respectfully disagrees with the above mentioned. In fact, those differing features have synergy because they both relate to fault detection efficiency and its improvement. When both sound and image are analysed, and in the case of fault (malfunction or predetermined deviation), the faults are presented with image and/or sound. Especially when they both are displayed to the user, it provides a more comprehensive knowledge/understanding of the fault, allowing it to be identified even more effectively, and then enabling correcting of the fault.

The features of the amended claim 1 differing from D1 provide the technical effect of a more comprehensive fault detection system that effectively aids in fault localization. Thus, starting from the D1, the technical problem to be solved is how to provide an alternative comprehensive fault detection system that effectively aids in fault localization and/or recognition. And the solution is stated above.

When looking at D1 and considering how to solve the objective technical problem of how to provide an alternative comprehensive fault detection system that effectively aids in fault localization and/or recognition, the skilled person would not find solution to the problem, but the idea of D1 is on an area monitoring system that combines real-time video from cameras with a rendered virtual environment, allowing a user to navigate the virtual environment and see the live video overlaid in the appropriate locations. There is no mention of fault detection, fault localization, or fault recognition capabilities in D1. The system described in D1 is intended for general area monitoring and surveillance, not specifically for fault detection or diagnosis. D1 does not provide a solution to this problem and at least an alarming system of the present invention is not disclosed.

When looking at D1 and considering how to solve the objective technical problem of how to provide an alternative comprehensive fault detection system that effectively aids in fault localization and/or recognition, the skilled person may look at D2. D2 discloses a system, which has a capability to analyze audio data, in addition to video data, to detect potential issues or maintenance problems based on recognizing characteristic sounds from the industrial equipment. When an issue or event that requires attention is detected, a notification is sent to the personnel. But the notification does not give such an information of the issue or event as the

present invention. At least an alarming system of the present invention is not disclosed. Based on previous a combination of D1 and D2 would not teach a skilled person to arrive at the claims.

When looking at D1 and considering how to solve the objective technical problem of how to provide an alternative comprehensive fault detection system that effectively aids in fault localization and/or recognition, the skilled person may look at D3. D3 is focused on a method for overlaying virtual image and audio data onto a real-time optical and acoustic reproduction of a real-world scene captured by a mobile device's sensors. There is no mention of fault detection, localization, or recognition in D3. The technology described is focused on augmented reality applications, not fault diagnosis systems. At least an alarming system of the present invention is not disclosed. Based on previous a combination of D1 and D3 would not teach a skilled person to arrive at the claims.

When looking at D1 and considering how to solve the objective technical problem of how to provide an alternative comprehensive fault detection system that effectively aids in fault localization and/or recognition, the skilled person may look at D4. The idea of D4 is on segmenting a person or object from an image and replacing it with different content to create a composite image. It does not mention anything about fault detection or fault localization systems. At least an alarming system of the present invention is not disclosed. Based on previous a combination of D1 and D4 would not teach a skilled person to arrive at the claims.

Based on previous a combination of D1 and D2, D3 or D4 would not teach a skilled person to arrive at the claims disclosing a comprehensive fault detection system that effectively aids in fault localization and/or recognition.

Consequently, the independent claims 1, 9 and 13 involve an inventive step over the cited prior art in sense of Art. 56 EPC.

**Conclusion**

Based on the above it is assumed that the application is now in order for allowance. If the examiner is of the opinion that further objections should be raised, then a further written communication is requested. In the event that the examiner considers that any further objections cannot be dealt with in writing, then oral proceedings are requested in precaution under Art. 116(1) EPC.

We reserve the right to restore and/or file a divisional application for any subject matter which might otherwise be regarded as abandoned or effectively disclaimed by virtue of the amendments now made.

In case you want to discuss this matter, please do not hesitate to contact Pauliina Wehkamp, Tel. +358 10 227 2639 or Pauliina.wehkamp@berggren.fi.

**Berggren Oy**

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Encl. Amended set of claims (marked-up and clean versions)
Amended pages of the description (marked-up and clean versions)